

COMPOSITE MATERIAL DATASHEETS CFRP AS4/8552-PW

MATERIAL IDENTIFICATION

EU 1E001 - This data is controlled under EU Dual Use Regulation 2021/821, as amended. Authorization may be required for the export of the data outside of the EU.

Short informal name	AS4/8552-PW
Commercial name	Hexcel Hexply AGP-193-PW/8552 RC40
Specifications	ABS5003 A-03 (AIMS05-01-003)
Description	CFRP Plain-Weave Fabric epoxy prepreg / 180°C Curing Class, for hand lay-up production
Version	0.1

REFERENCES

- TEAS-NT-1910014 Is. H (NT-DVs-F10X) , AS4-8552-PW_70C_legacy_values_forced.dat/out

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.205 mm
Density	1560 kg/m ³
Tg-onset (wet)	150 °C
Max. Operational Limit (MOL)	120 °C

Coefficients of Thermal Expansion CTE (1/°C)	Range	-55...20°C	20...70°C	70...120°C
	α_1	2.7E-6	1.8E-6	1.8E-6
	α_2	2.7E-6	1.8E-6	1.8E-6
	α_3	4.0E-5	4.7E-5	5.3E-5

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12}	K_B
Values	58000	58000	3500	0.05	0.86

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1 , E_2 and G_{12} in buckling stability analyses

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13}	ν_{23}
Values	11500	3200	3200	0.48	0.48

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
F_{1t}	-	708.7	708.7	708.7	0.889	-	1	1	-	630	630	630
F_{1c}	813.5	818.1	652.4	599.7	0.769	0.994	0.797	0.733	625.6	629.2	501.7	461.1
F_{2t}	-	708.7	708.7	708.7	0.889	-	1	1	-	630	630	630
F_{2c}	813.5	818.1	652.4	599.7	0.769	0.994	0.797	0.733	625.6	629.2	501.7	461.1
F_{12}	120.7	124	97.4	84.3	0.9	0.973	0.786	0.68	108.6	111.6	87.7	75.9

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values ($\mu\epsilon$)				B-basis K_B	E-KDF			B-basis Values ($\mu\epsilon$)			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
ϵ_{tall}	-	11500	11500	11500	0.889	-	1	1	-	10200	10200	10200
ϵ_{call}	14000	14100	11200	10300	0.769	0.993	0.794	0.73	10800	10800	8600	7900

COMPOSITE MATERIAL DATASHEETS

CFRP AS4/8552-PW

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
F_{13} (MPa)	-	76.7	55.9	48.6	0.866	-	0.729	0.634	-	66.4	48.4	42.1
F_{3t} (MPa)	-	74.4(*)	33.6(*)	33.6(*)	0.866	-	0.452	0.452	-	64.4	29.1	29.1
G_{IC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-

(*) Through-thickness tensile strength F_{3t} for this fabric material has been made equal to the F_{2t} strength of the UD material of the same family Hexcel AS4/8552-UD (data from references QTR150635, NT-F-ID-99102)

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	-55C/Dry	RT/Dry	70°C/W	90°C/W		-55C/Dry	70°C/W	90°C/W	-55C/Dry	RT/Dry	70°C/W	90°C/W
OHT_{QI}	-	268.8	268.8	268.8	0.9	-	1	1	-	241.9	241.9	241.9
OHC_{QI}	-	318	267.4	245	0.9	-	0.841	0.770	-	286.2	240.7	220.5
FHT_{QI}	361.2	271.8(**)	!	!	0.888	>1	1	1	320.7	241	241	241
FHC_{QI}	-	364.3	364.3	364.3	0.814	-	1	1	-	296.6	296.6	296.6
F_{BR-QI}	-	619.7	487.3	462.2	0.9	-	0.786	0.746	-	557.7	438.6	416
F_{PO}	-	-	-	-	-	-	-	-	-	-	-	-

(**) Datum not obtained from material qualification, but from a stress allowables tests campaign (NT-00-ECE-017), in this case tested at cold ambient, which is conservative for this property

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

Property	B-basis Value for all conditions
TAI ($\mu\epsilon$)	4600
CAI ($\mu\epsilon$)	2200

$$E/t^2 = A \cdot (1 - e^{-B \cdot d_0/t})$$

Approximation model type: SMM chapter 1421: CDTM-T05

$$CAI = K_B \cdot EKDF \cdot \{ CAI_{\infty-QI} \cdot (3 / K_{TOH})^{C_x} + A / e^{B \cdot e \cdot (E/t^2)} \}$$

Property	Design Value
A (J/mm ²)	
B	

Property	Mean Value RT/Dry	B-basis K_B	E-KDF		
			70°C/W	90°C/W	120°C/W
$CAI_{\infty-QI}$ ($\mu\epsilon$)					
A ($\mu\epsilon$)					
B_e					
C_x					

...where:

- E is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- t the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

COMPOSITE MATERIAL DATASHEETS CFRP AS4/8552-PW

$$K_{TOH} = 1 + \{ 2 \cdot [(E_x/E_y)^{0.5} - \nu_{xy}] + E_{xy}/G_{xy} \}^{0.5}$$

(*) CAI shall be calculated not only for the 0° direction (X parallel to material 0° reference), but also for the rest of ply orientations, i.e. the 45° and 90° directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x , E_y , G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the allowable strains applicable to plies at 0° (CAI_0), at ±45° (CAI_{45}) and at 90° (CAI_{90}).

DRAFT